Task S3.01. Manipulation of tables

Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

Level 1

Exercise 1

Your task is to design and create a table called "credit_card" that stores crucial details about credit cards. The new table must be able to uniquely identify each card and establish an appropriate relationship with the other two tables ("transaction" and "company"). After creating the table, you will need to enter the information in the document called "dades_introduir_credit". Remember to show the diagram and provide a brief description of it.

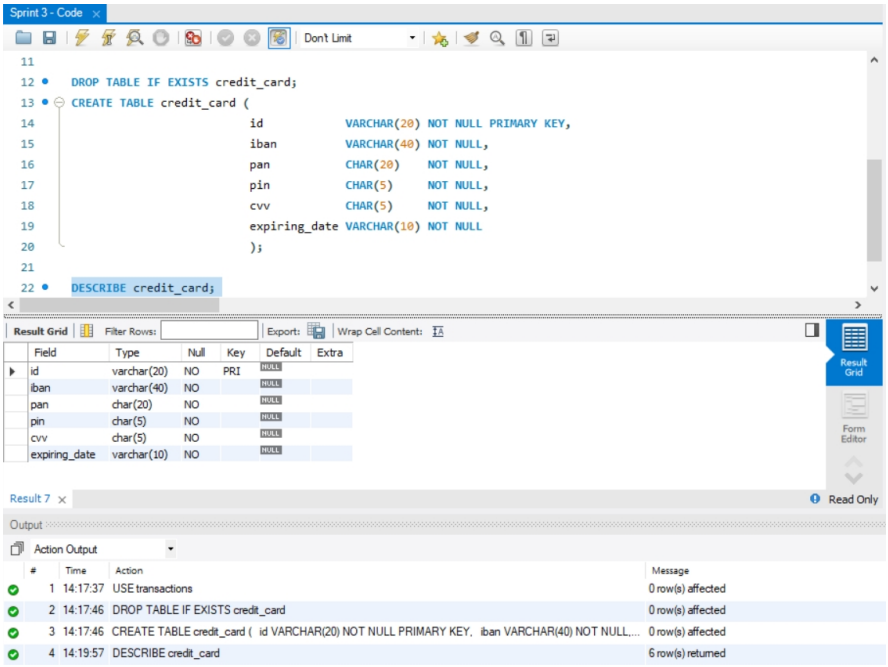


Fig 1. *Creating the* `credit_card` *table and checking its parameters*

Once the table `credit_card` is created, we can copy the data from datos_introducir_sprint3_credit file and paste it into a New Query Tab to import it into our existing table.

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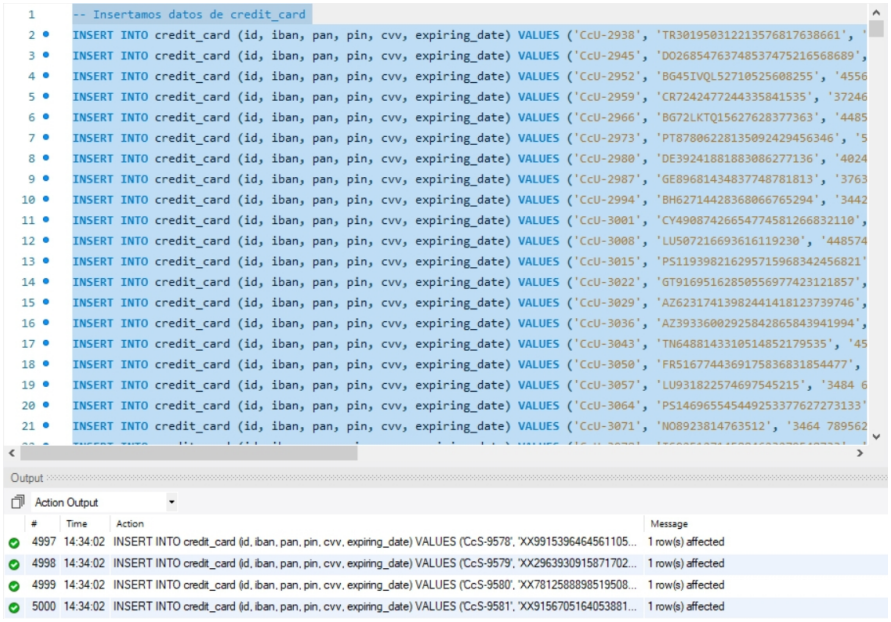


Fig 2. *Inserting (importing) data to the* `credit_card` *table*

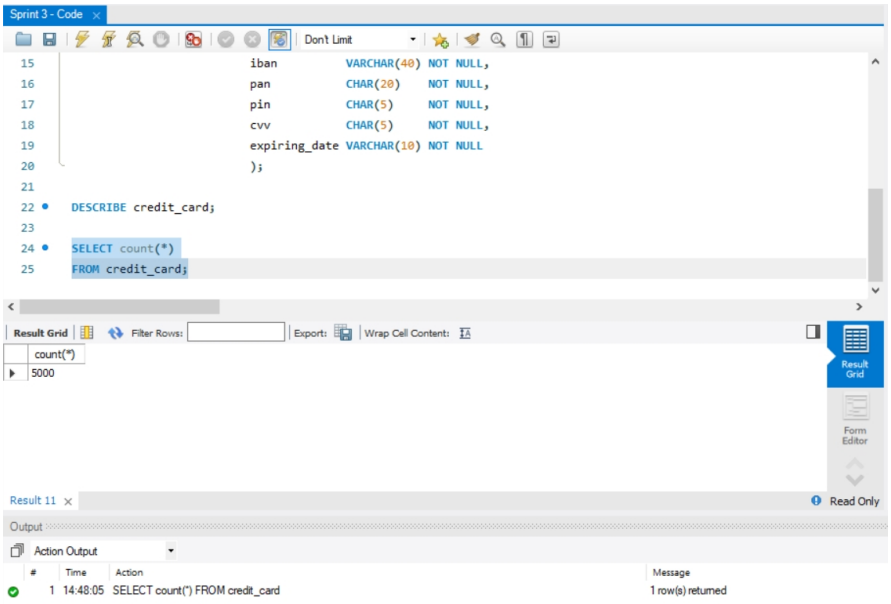


Fig 3. *Imported data check*

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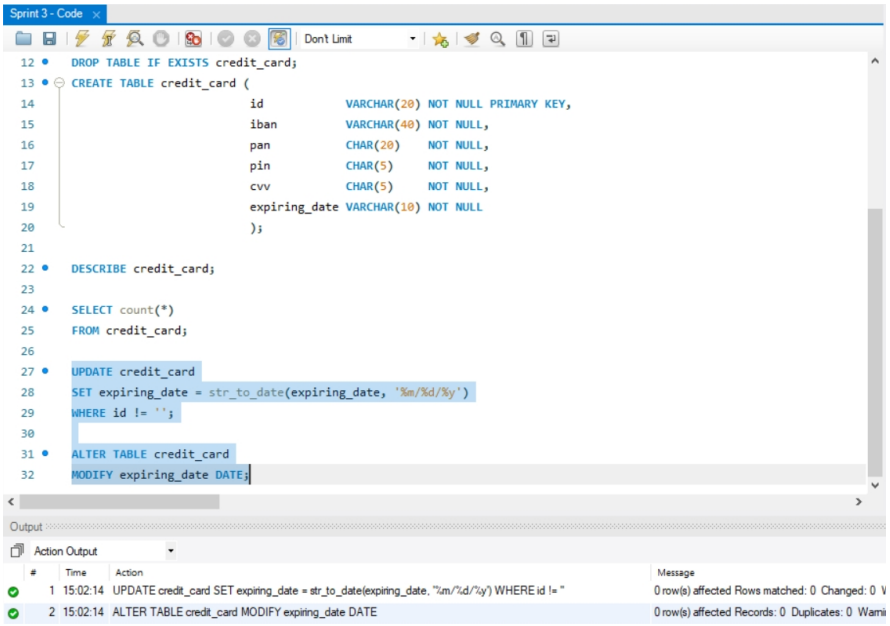


Fig 4. *Ensuring the correct date format in the table*

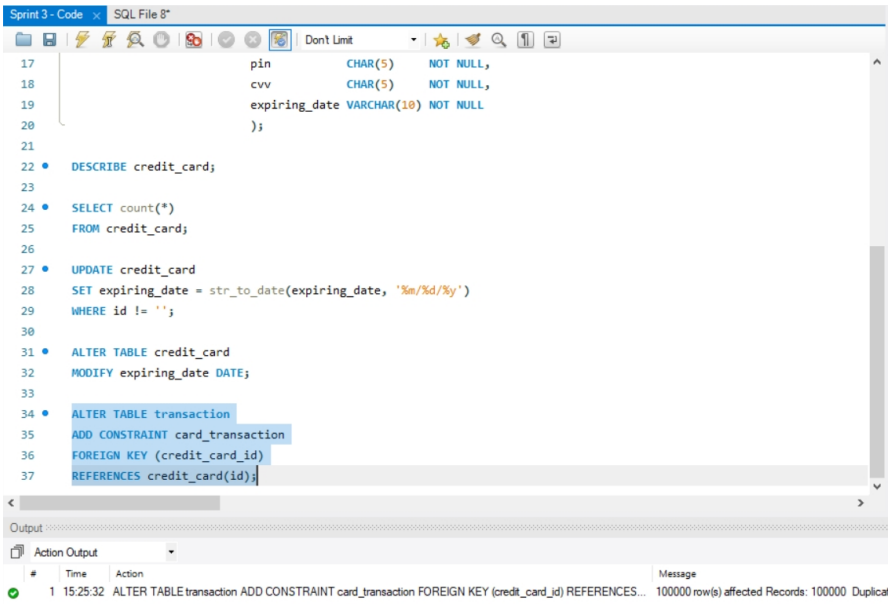


Fig 5. *Connecting two tables using PK-FK*

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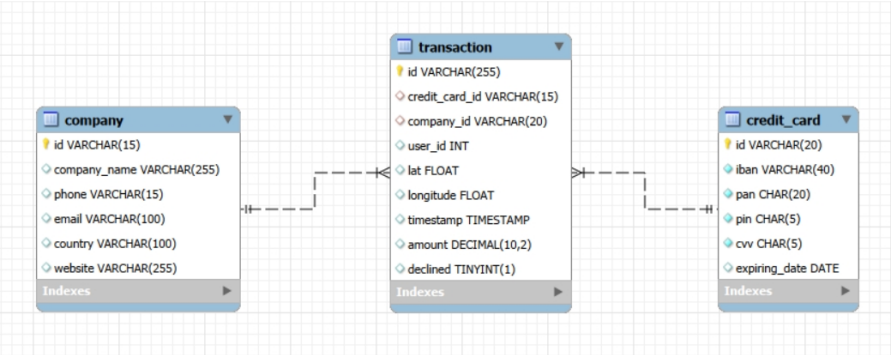


Fig 6. *ER diagram after adding the* `credit_card` *table*

Figure 6 shows the ER diagram for the "transactions" database. This database, which has a simple star-type relational model now includes three tables: `company` (a dimensional table), `credit_card` (another dimensional table), and `transaction` (a fact table). Both the `company` table and the `credit_card` table are related to the `transaction` table via a 1:N granularity by the primary key "id" in the `company/credit_card` table to the foreign key "company_id/credit_card_id" in the `transaction` table (which has its own primary key: "id"). The ER diagram also illustrates the column names and their corresponding data types for each table.

Exercise 2

The Human Resources department has identified an error in the account number associated with the credit card with ID CcU-2938. The information that should be displayed for this record is: TR323456312213576817699999. Remember to show that the change was made.

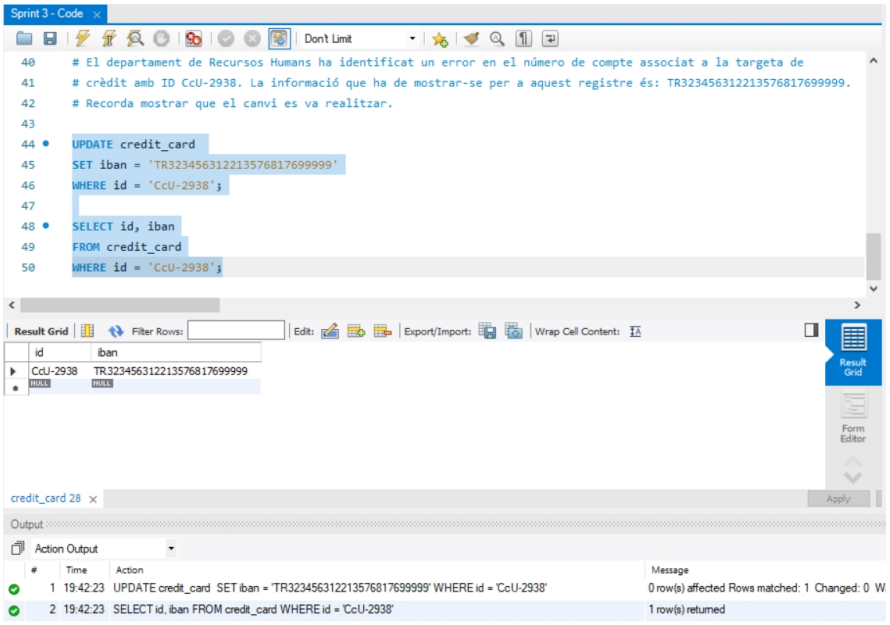


Fig 7. *IBAN change for a specific ID*

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Exercise 3

In the "transaction" table, enter a new transaction with the following information:

Id	108B1D1D-5B23-A76C-55EF-C568E49A99DD
credit_card_id	CcU-9999
company_id	b-9999
user_id	9999
lat	829.999
longitude	-117.999
amount	111.11
declined	0

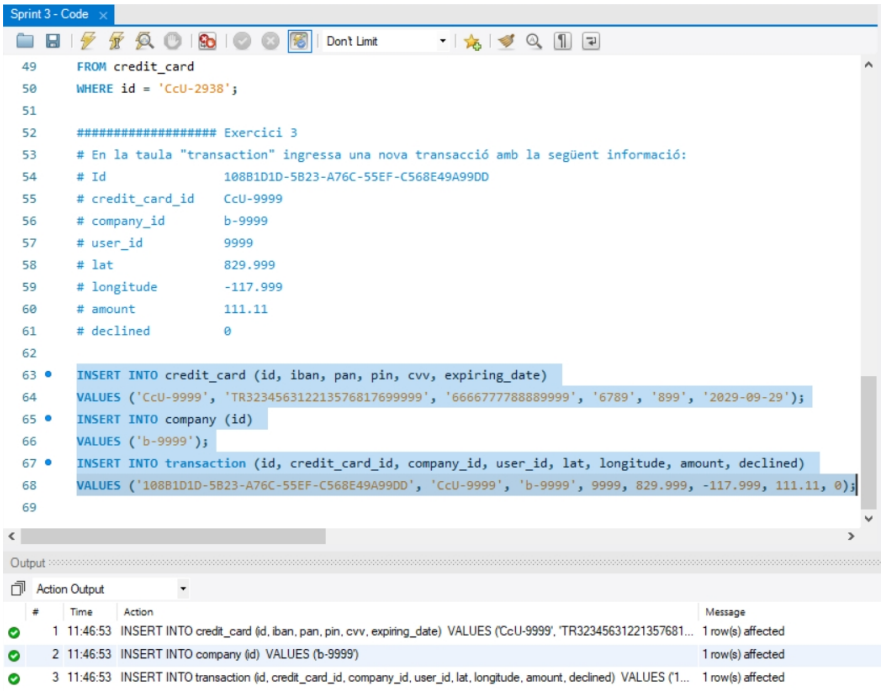


Fig 8. *Adding a new transaction*

Since the `transaction` table is connected to the `company` and `credit_card` tables via foreign keys, adding a new row to the `transaction` table is not sufficient; the corresponding data must also be added to the `company` and `credit_card` tables. Moreover, since we used NOT NULL to avoid NULL values in creating the `credit_card` table, default values are needed to link the added row to the `transaction` table.

Exercise 4

Human resources have asked you to delete the "pan" column from the credit_card table. Remember to show the change made.

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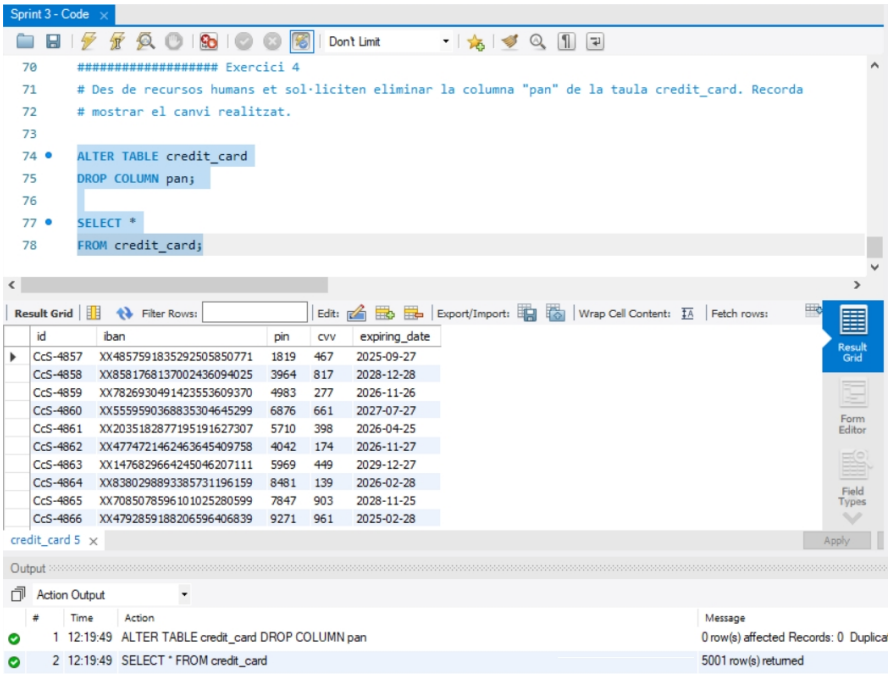


Fig 9. *Eliminating a column (pan)*

Level 2

Exercise 1

Remove from the transaction table the record with ID 000447FE-B650-4DCF-85DE-C7ED0EE1CAAD from the database.

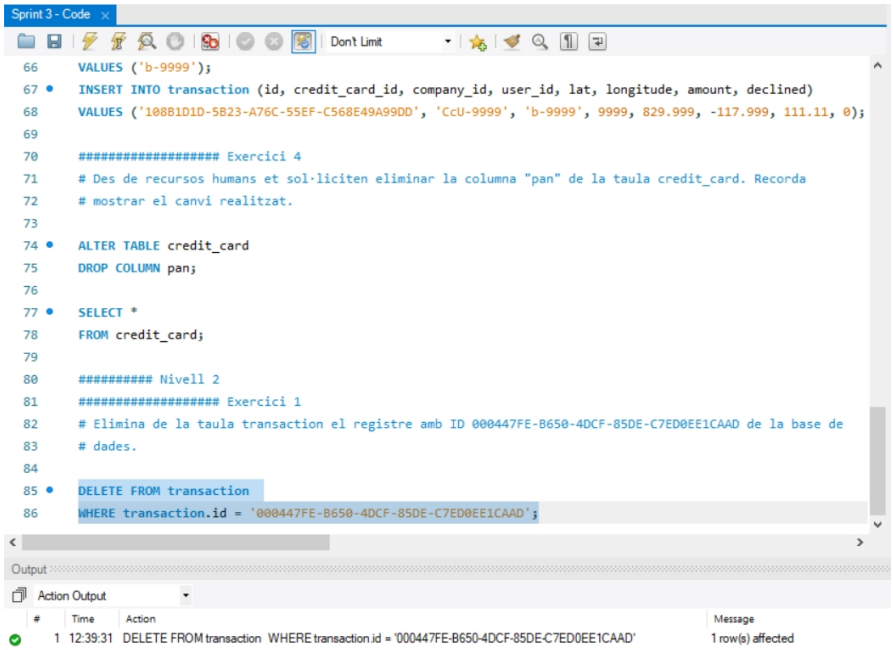


Fig 10. *Removing a specific transaction*

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Exercise 2

The marketing department wants to have access to specific information to perform effective analysis and strategies. It has been requested to create a view that provides key details about the companies and their transactions. You will need to create a view called VistaMarketing that contains the following information: Company name, Contact phone number, Country of residence, and Average purchase made by each company. Present the view created, ordering the data from highest to lowest average purchase.

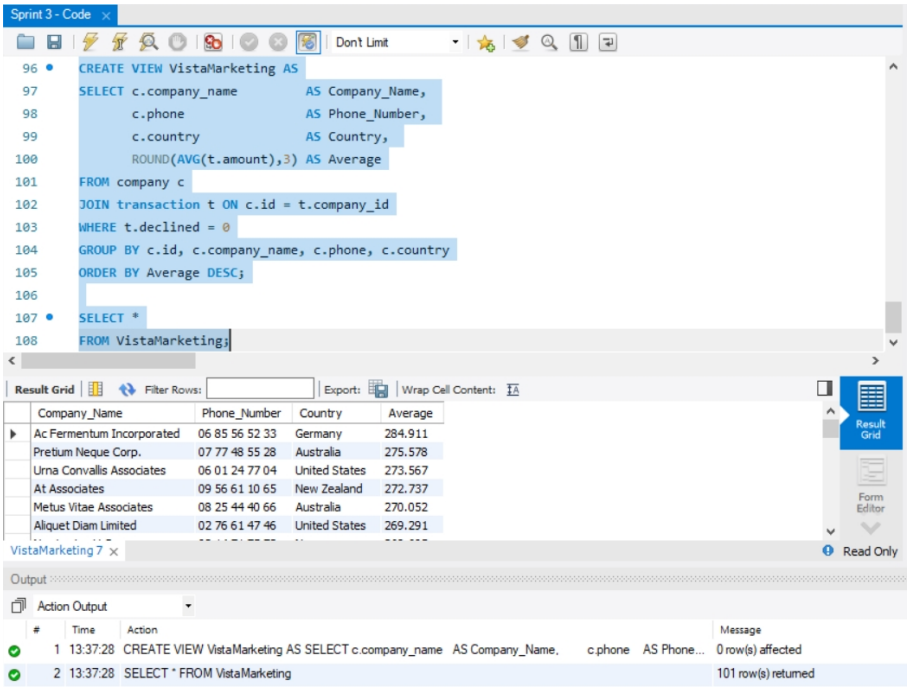


Fig 11. *Creating the VIEW requested by the marketing department*

Exercise 3

Filter the VistaMarketing view to show only companies that have their country of residence in "Germany"

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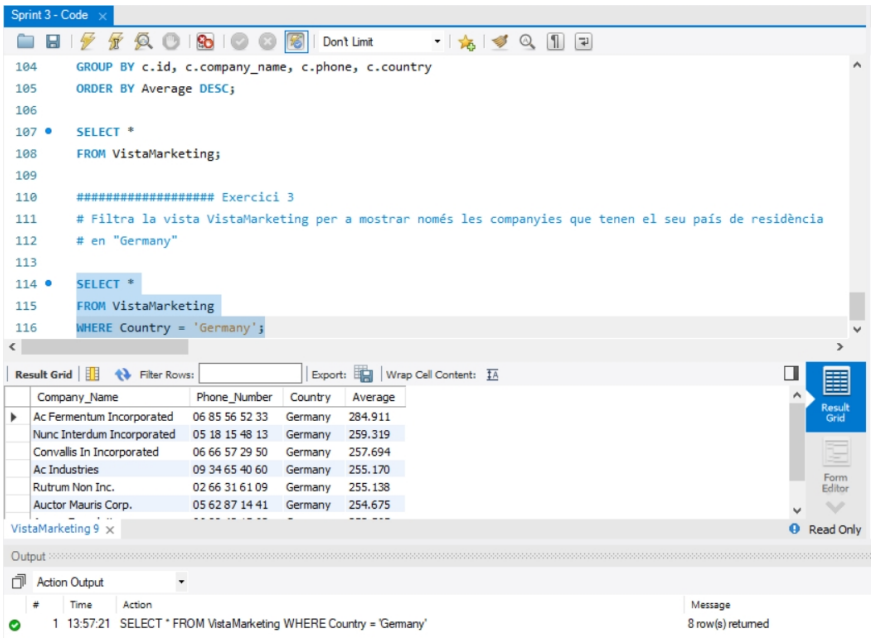
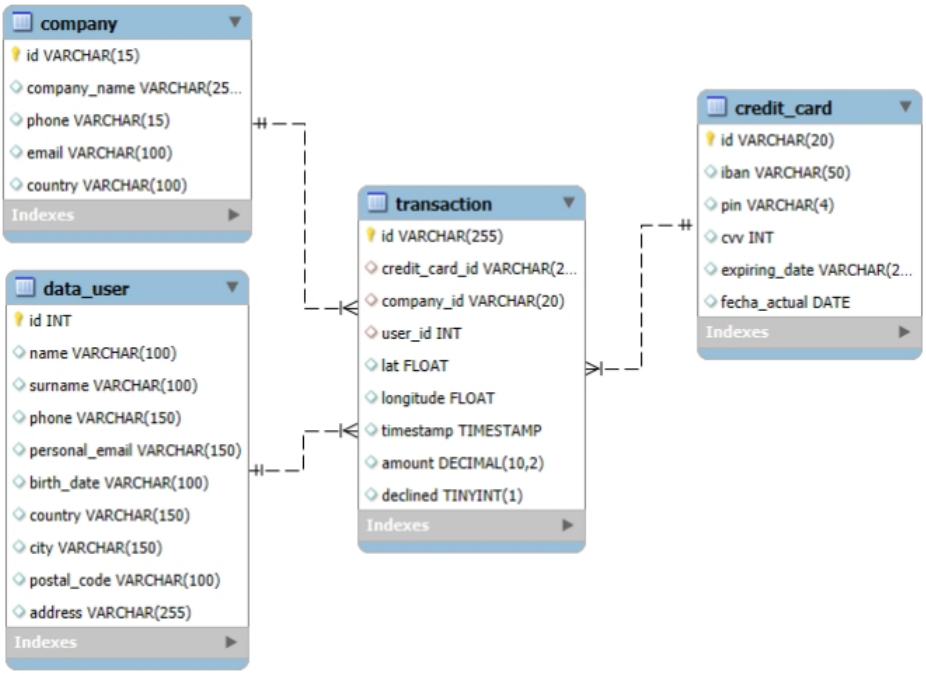


Fig 12. *Filtering a VIEW*

Level 3

Exercise 1

Next week you will have another meeting with the marketing managers. A colleague of your team made modifications to the database, but he does not remember how he did them. He asks you to help him leave the commands executed to obtain the following diagram:



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First, we need to run "estructura datos user" to create the user table and then run "datos introducir sprint3 user" to insert data.



Fig 13. *Creating the user table*

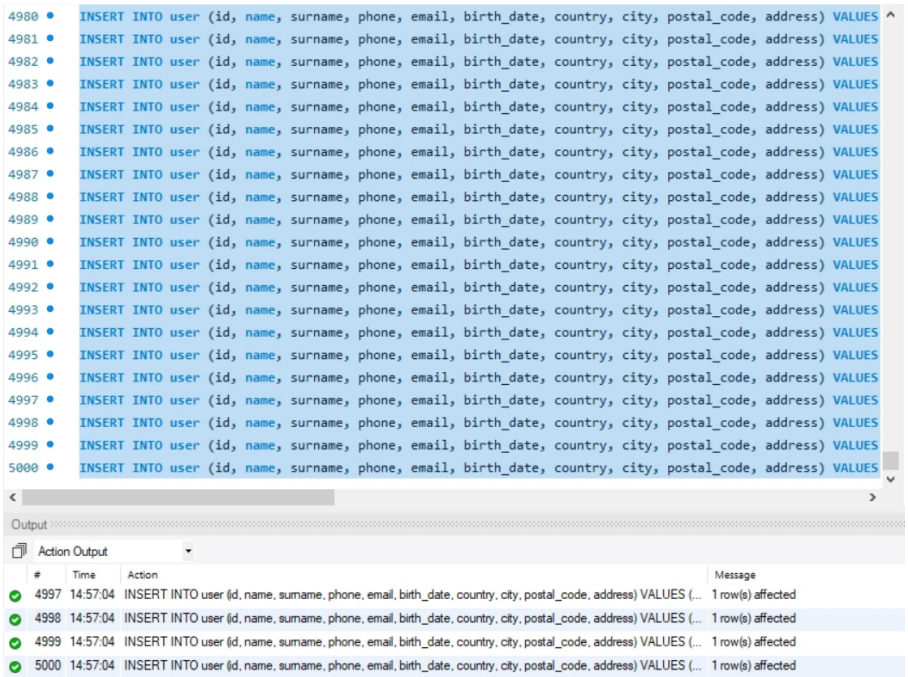


Fig 14. *Inserting the data into the table*

Each table needs some changes to maintain the schematic model of this exercise. These changes will be discussed for each table.

user table:

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- Changing the table name to `data_user`
- Converting “id” type from CHAR(10) to INT
- Changing “email” to “personal_email”

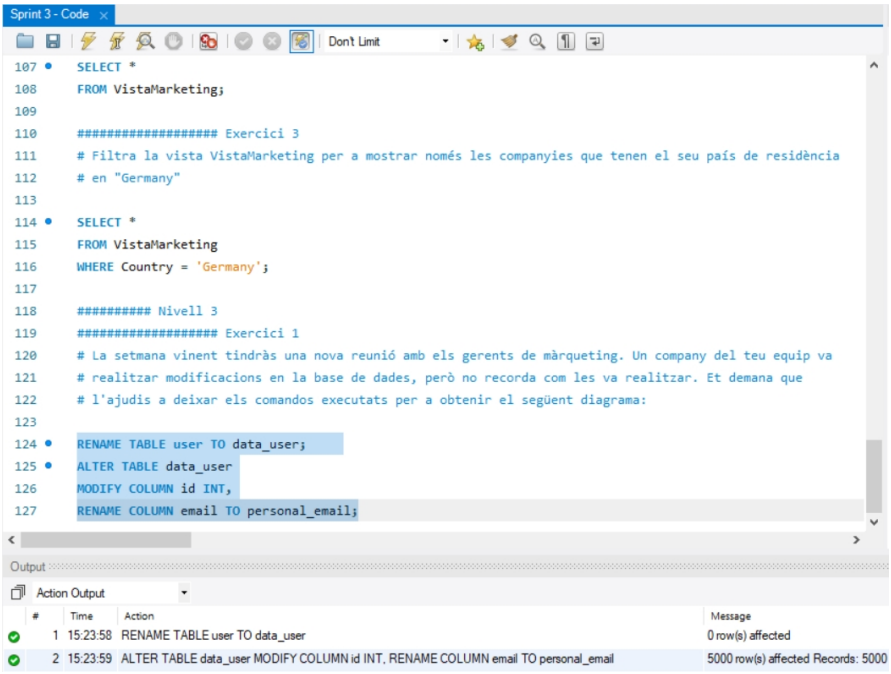


Fig 15. *Changes applied to the table user*

company table:

- Omitting the column “website”

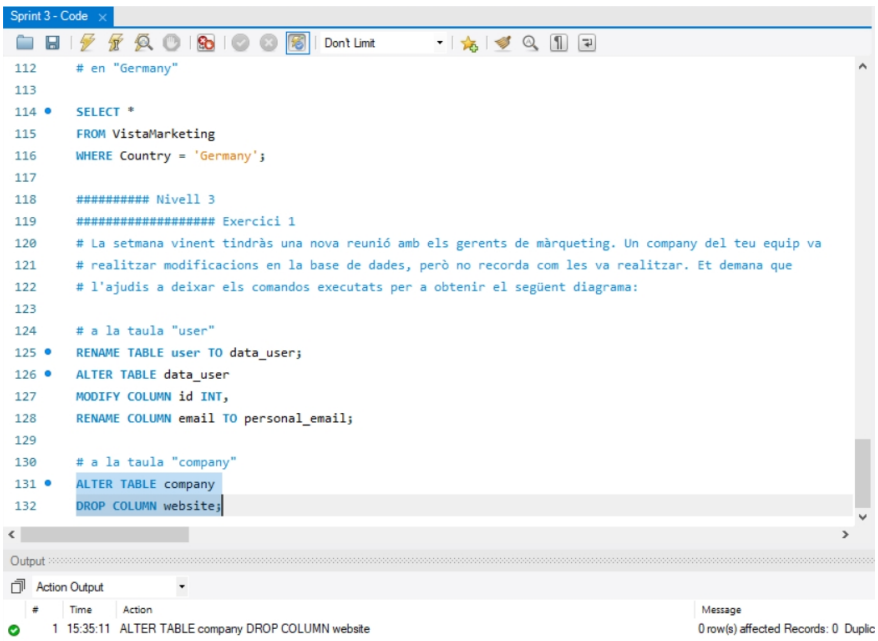


Fig 16. *Changes applied to the table company*

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`transaction` table:

- Converting “credit_card_id” type from VARCHAR(15) to VARCHAR(20)

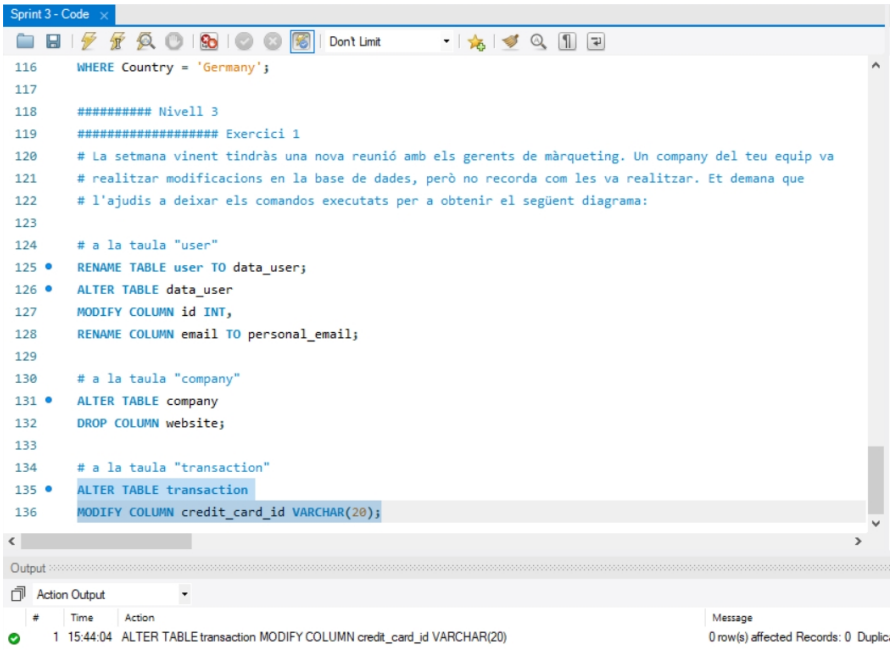


Fig 17. *Changes applied to the table* `transaction`

`Credit_card` table:

- Converting “iban” type from VARCHAR(40) to VARCHAR(50)
- Converting “pin” type from VARCHAR(5) to VARCHAR(4)
- Converting “cvv” type from VARCHAR(5) to INT
- Converting “expiring_date” type from DATE to VARCHAR(20)
- Adding “fecha_actual” column with DATE type

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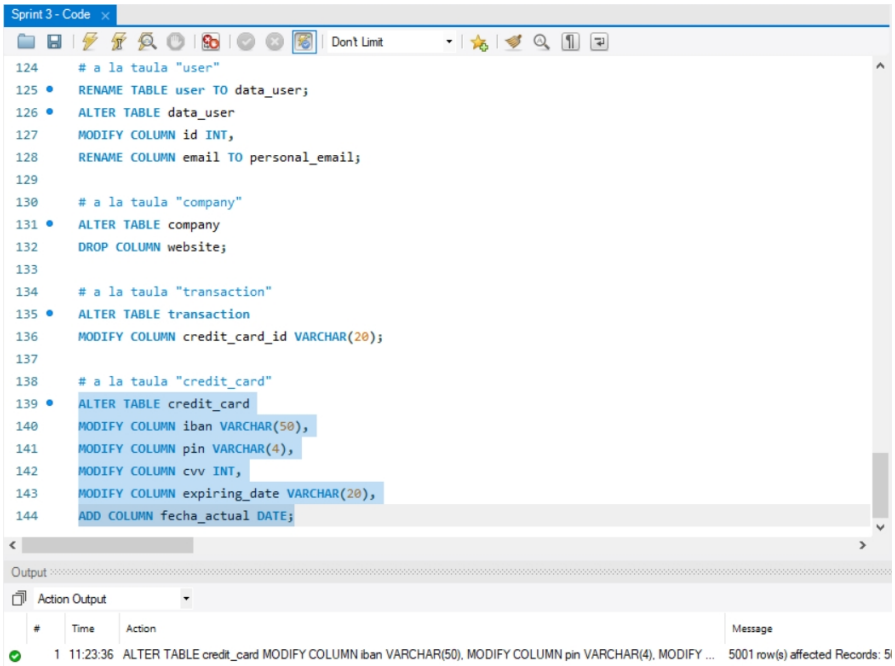


Fig 18. *Changes applied to the table* `credit_card`

Before connecting the new table (`data_user`) to our model, we need to check for unassigned (orphaned) data in `data_user` and assign the corresponding records from the `transaction` table.

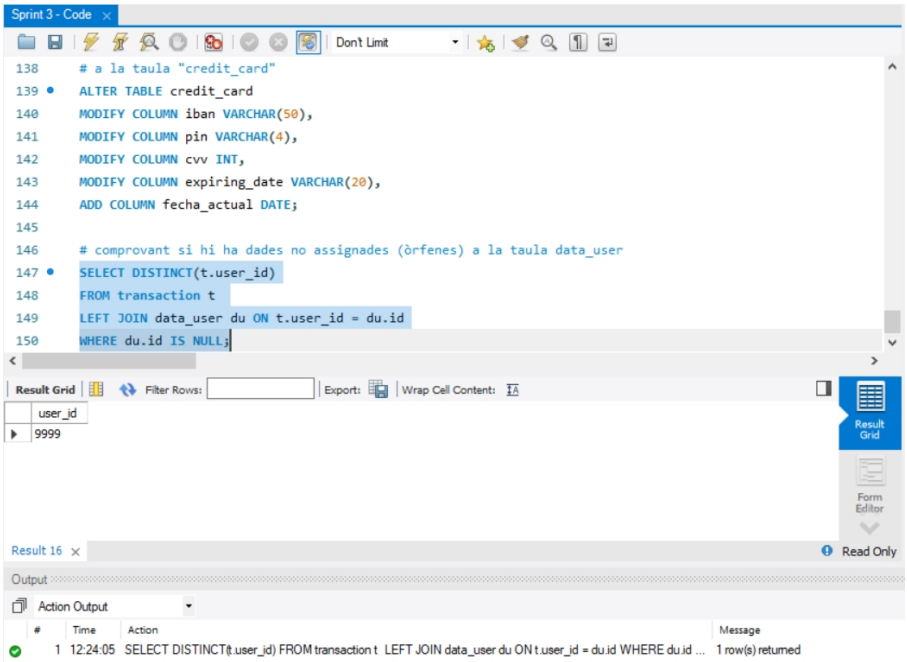


Fig 19. *Checking for unassigned (orphaned) data in the* `data_user` *table*

Task S3.01. Manipulation of tables

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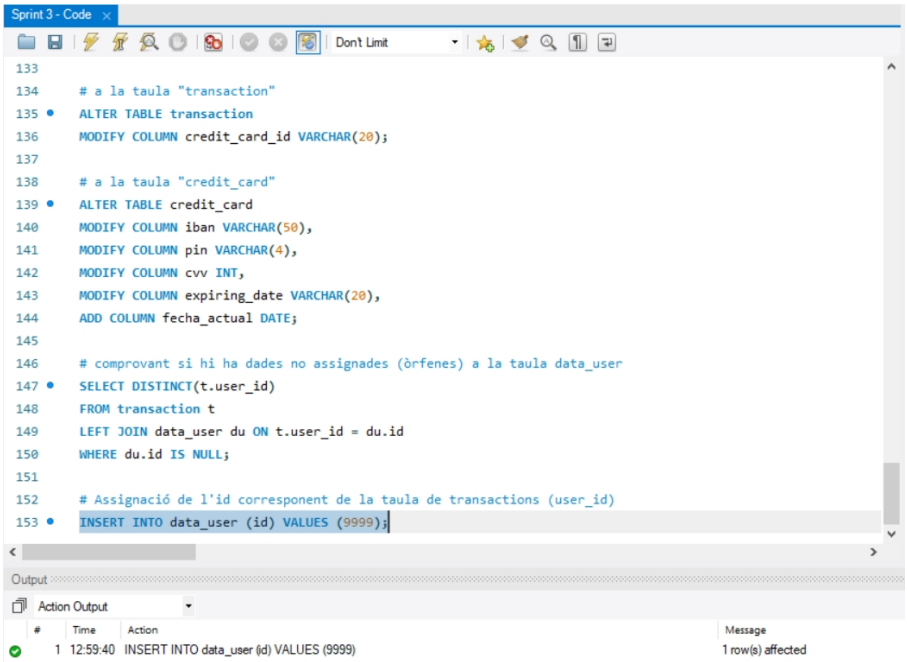


Fig 20. *Assignment of the corresponding id from the* `transaction` *table (user_id)*

Finally, we connect the `data_user` table using its Primary Key (id) to the `transaction` table with the Foreign Key (user_id).

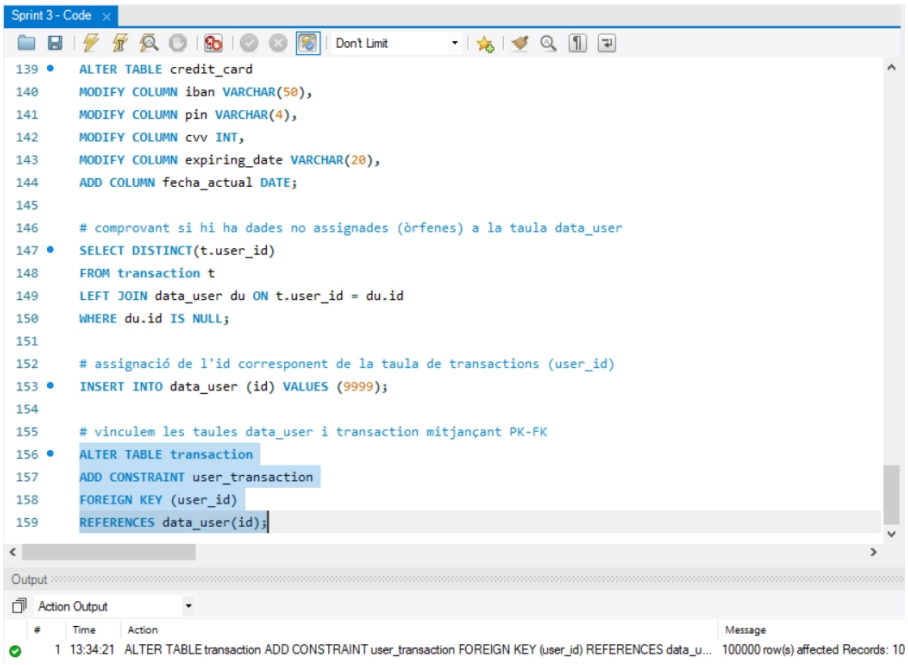


Fig 21. *Linking* `data-user` *and* `transaction` *tables via PK-FK*

As of the final step, we can compare the ER diagram of our database (Fig 22) with the one mentioned in the exercise. It can be seen that they are identical.

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Mani Rezaeirad *p2p review partner: Timan (Nakta) Samipour*

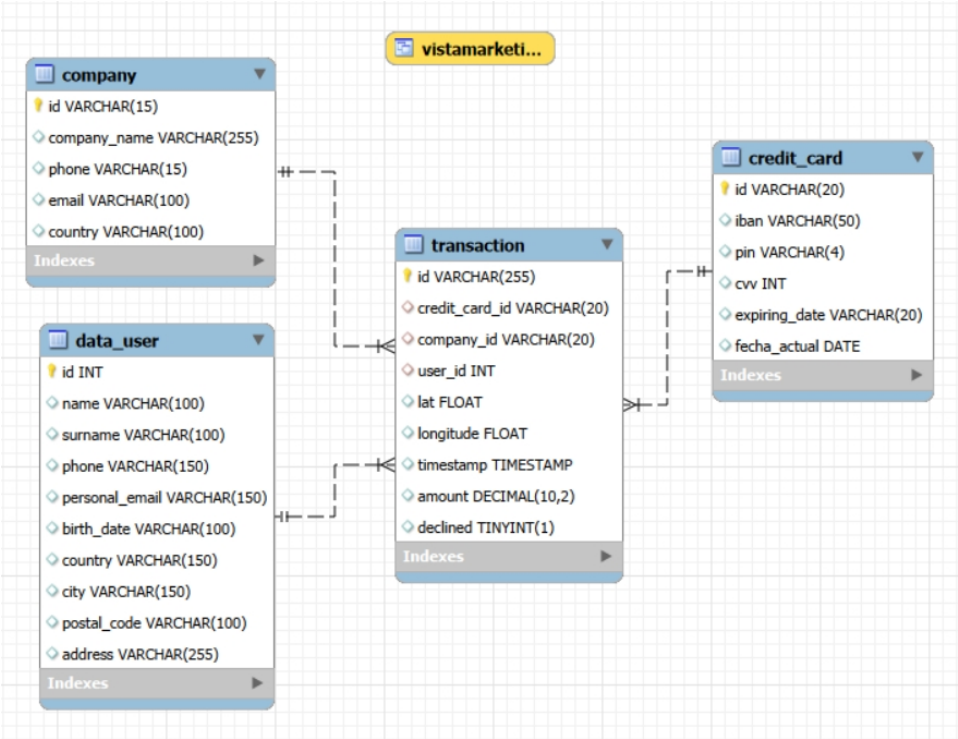


Fig 22. *ER diagram after adding and connecting the data_user table*

Exercise 2

The company also asks you to create a view called "InformeTecnico" that contains the following information:

- Transaction ID
- Username
- User's last name
- IBAN of the credit card used.
- Name of the company of the transaction carried out.
- Be sure to include relevant information from tables you will be familiar with, and use aliases to rename columns as needed.

Displays the results of the view, sorts the results in descending order based on the transaction ID variable.

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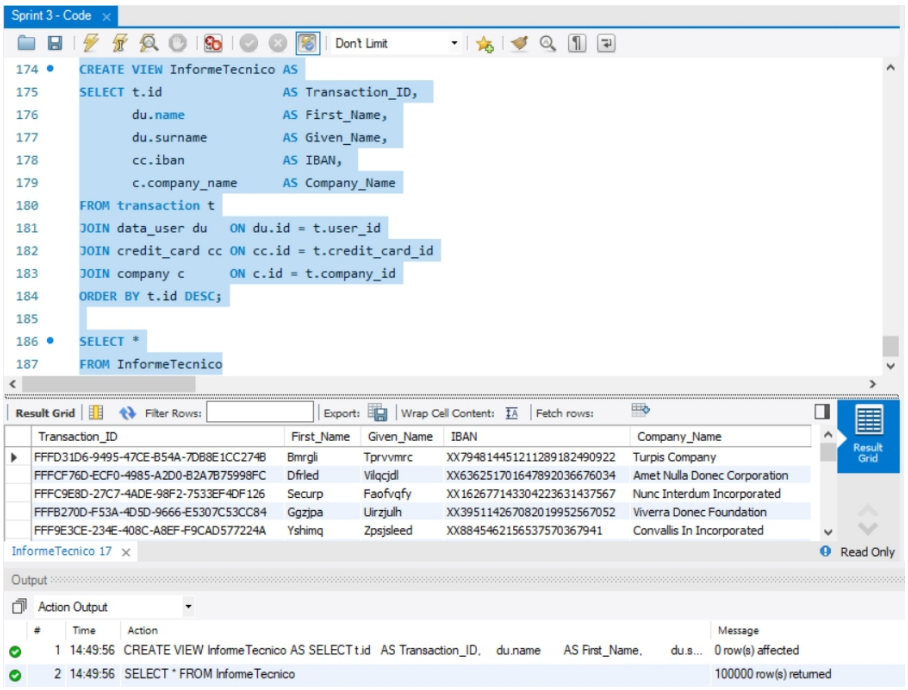


Fig 23. *Creating the VIEW "InformeTecnico" with the requested information*